Operating Systems Lecture 1: Process Management

Process Control Blocks and CPU Scheduling

A process is an active program in execution possessing its own distinct virtual address space and call stack.
The operating system kernel maintains a Process Control Block PCB tracking registers, program counter, and state.
Context switching saves hardware registers of the current process and loads the saved state of the next task.

Operating Systems Lecture 2: Memory Hierarchy and Paging

Virtual Address Translation and Page Replacement

Paging divides physical memory into fixed-size frames and logical process space into equal-sized virtual pages.
The memory management unit MMU uses translation lookaside buffers TLB to accelerate virtual-to-physical addres
When an accessed page is not resident in physical RAM, the MMU raises a page fault trap to fetch it from disk.

Operating Systems Lecture 3: Concurrency and Synchronization

Mutual Exclusion Primitives and Deadlock Prevention

Race conditions occur when multiple concurrent threads read and mutate shared mutable state without coordination
Mutual exclusion locks and counting semaphores ensure that only one thread executes inside the critical section.
Deadlocks require four simultaneous conditions: mutual exclusion, hold and wait, no preemption, and circular wait.